Words or Vision: Do Vision-Language Models Have Blind Faith in Text?

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Abstract

Vision-language models (VLMs) are increasingly deployed in pipelines where supplementary text—from retrieval-augmented generation, web agents, or OCR—may be noisy, irrelevant, or outright incorrect. Yet existing benchmarks rarely test how models arbitrate between conflicting visual and textual evidence. We introduce ***BFiT*** *(Blind Faith in Text), a diagnostic benchmark that augments four vision-centric tasks—VQAv2, DocVQA, MathVista, and Brand Recognition—with three controlled text conditions (matched, corrupted, and irrelevant), producing 16,500 evaluation samples from 5,500 base questions. Evaluating ten VLMs spanning proprietary and open-weight families, we find that text bias is pervasive: open-weight models frequently follow corrupted text over clear visual evidence (e.g., LLaVA-NeXT-7B reaches a Text Preference Ratio of 85.52% on VQAv2), whereas proprietary models exhibit substantially greater robustness. We further analyze factors driving this bias—including language-model scale, token ordering, text relevance, and cautionary instructions—and demonstrate that supervised fine-tuning with text-augmented data substantially reduces over-reliance on text while generalizing across tasks.*

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1. Introduction

Vision-language models (VLMs) have achieved remarkable progress on tasks ranging from visual question answering [12, 23] to document understanding [28] and mathematical reasoning [27]. As these models move from controlled benchmarks into real-world deployments, they increasingly operate within larger systems—retrieval-augmented generation (RAG) pipelines [19, 30], autonomous web agents [17, 49], and multimodal assistants [9]—that supply additional textual context alongside images. In such settings, the accompanying text may be noisy, outdated, irrelevant, or even adversarially manipulated [34, 42]. A reliable VLM must therefore be able to reconcile conflicting signals across modalities, trusting visual evidence when the text is misleading.

However, existing multimodal benchmarks [20, 26, 46] evaluate VLMs primarily in settings where all input modalities are consistent and cooperative. They do not stress-test the critical failure mode in which textual and visual information disagree. This gap leaves open a fundamental question: *when text conflicts with an image, do VLMs faithfully interpret what they see, or do they exhibit blind faith in text?*

To answer this question, we introduce the **Blind Faith in Text (BFiT)** benchmark. BFiT augments four established vision-centric tasks—VQAv2 [12] (1k samples), DocVQA [28] (1k), MathVista [27] (1k), and Brand Recognition [22] (2.5k)—with three controlled text conditions: *matched* text that supports the correct answer, *corrupted* text that plausibly points to an incorrect answer, and *irrelevant* text drawn from unrelated Wikipedia passages [29]. This design yields 16,500 evaluation samples from 5,500 base questions and enables fine-grained measurement of modality preference through a novel *Text Preference Ratio* (TPR) metric.

We evaluate ten VLMs spanning proprietary systems (GPT-4o [2], GPT-4o mini, Claude Sonnet, Claude Haiku [3]) and open-weight models (LLaVA-NeXT 7B/13B/34B [24], Phi3.5 [1], Molmo-7B-D [7], Qwen2-VL-7B [40]). Our findings reveal a striking pattern: open-weight models frequently exhibit "blind faith in text," preferring corrupted textual cues over clear visual evidence (e.g., LLaVA-NeXT-7B TPR reaches 85.52% on VQAv2 under corruption). Proprietary models are substantially more robust, with Claude Sonnet achieving a TPR of only 9.58% on VQAv2. We further investigate how several factors modulate text bias, including language-model backbone size, token ordering (text before vs. after image), text relevance, and explicit cautionary instructions. While each factor has measurable effects, none fully eliminates the bias.

Finally, we demonstrate that supervised fine-tuning (SFT) with text-augmented training data offers a practical and effective mitigation strategy. Fine-tuned models show dramatic improvements—LLaVA-7B corruption accuracy improves from 28.69% to 71.25%, and Qwen2-VL-7B from 50.79% to 82.72%—with gains that generalize to held-out tasks. We further provide a theoretical analysis linking text bias to the imbalance between pure-text and multimodal data in pretraining.

Our main contributions are as follows:

- We introduce **BFiT**, a systematic benchmark for evaluating VLMs' modality arbitration under controlled text–

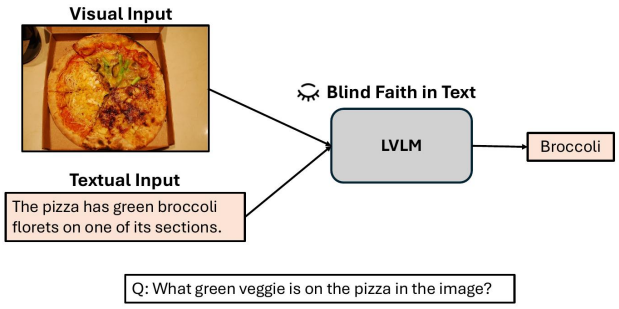

Figure 1. Illustration of the Blind Faith in Text phenomenon. When provided with textual context that conflicts with visual evidence, VLMs may follow the text to produce an incorrect answer.

image conflicts across four diverse vision tasks.

- We reveal and quantify a pervasive "blind faith in text" phenomenon, showing that open-weight VLMs suffer substantial accuracy degradation when presented with corrupted textual context.
- We diagnose factors driving text bias, including model scale, token order, text relevance, unimodal certainty, and cautionary instructions.
- We show that supervised fine-tuning with text-augmented data substantially reduces text bias and generalizes across tasks, offering a practical mitigation path.
- We provide a theoretical framework linking text over-reliance to training data composition, grounding our empirical findings.

2. The BFiT Benchmark

We introduce the **Blind Faith in Text (BFiT)** benchmark, designed to systematically evaluate whether VLMs overrely on textual context when it conflicts with visual evidence. Figure 1 illustrates the core phenomenon: given an image and a conflicting textual description, VLMs may follow the text rather than the image to answer the question. BFiT constructs controlled text variations around standard vision tasks and measures the resulting behavioral shifts across a diverse set of models.

2.1. Datasets

BFiT spans four vision-centric tasks of varying complexity:
VQAv2 [12]: We sample 1,000 questions from the validation set, covering general visual question answering over natural images. This dataset tests basic visual understanding, including object recognition, counting, and spatial reasoning.
DocVQA [28]: We sample 1,000 questions from the validation set, focusing on document understanding tasks involving charts, tables, and structured layouts. This evaluates whether models can extract information from visually grounded documents despite conflicting text.
MathVista [27]: We use 1,000 questions from the minitest split, requiring mathematical reasoning over visual inputs such as plots, geometric diagrams, and function graphs. This tests robustness of analytical reasoning under textual interference.
Brand Detection [22]: We use 2,500 samples from the KnowPhish test set, where each sample consists of a webpage screenshot paired with its HTML source. This real-world security task requires identifying brand logos and visual indicators, representing a high-stakes setting where text bias could have practical consequences.

Together, these four datasets yield 5,500 base questions spanning general perception, document comprehension, mathematical reasoning, and applied security.

2.2. Text Construction

For each base triplet (image, question, answer), we construct three textual conditions:
Matched text. Using GPT-4o [2], we generate a short textual description that provides sufficient information to answer the question correctly without seeing the image. This serves as a cooperative control where text and image are aligned.
Corrupted text. Again using GPT-4o, we generate a plausible but misleading description that, if trusted, would lead to a *different* answer than the ground truth. This is the key probe for text over-reliance: a model that follows corrupted text over clear visual evidence exhibits blind faith in text.
Irrelevant text. We sample unrelated passages from WikiText [29], drawn from Wikipedia articles with no connection to the image or question. This condition tests whether the mere presence of extraneous text degrades visual reasoning.

The three conditions applied to 5,500 base questions produce a total of 16,500 test samples. Each condition isolates a distinct aspect of modality interaction: alignment, conflict, and noise.

2.3. Models and Evaluation Protocol

We evaluate ten VLMs spanning proprietary and open-weight families, all run with temperature 0 for deterministic decoding:
Proprietary models: GPT-4o, GPT-4o mini [2], Claude Sonnet 3.5, Claude Haiku 3.5 [3].
Open-weight models: LLaVA-NeXT-7B, 13B, and 34B [24], Phi3.5 [1], Molmo-7B-D [7], Qwen2-VL-7B [40].

Each model receives an input of the form: `[image] + [optional cautionary instruction] + [textual context] + [question]`. We record model responses and classify each into one of three behavioral categories: *image-consistent* (answer matches the visual ground truth),

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text-consistent (answer matches the textual suggestion), or *other* (neither).

Metrics. We report the following:

- **Accuracy**: standard per-condition accuracy.
- **Macro Accuracy**: average accuracy over the matched, corrupted, and irrelevant conditions.
- **Normalized Accuracy**: corruption accuracy divided by base accuracy, scaled to 100. This measures relative robustness to text corruption independent of baseline capability.
- **Text Preference Ratio (TPR)**: the fraction of image-text conflict cases (under corruption) where the model follows the text rather than the image. TPR directly quantifies blind faith in text; lower is better.

2.4. Sanity Check

To validate that our constructed text conditions behave as intended, we conduct a text-only evaluation in which models receive only the textual context and question—without any image. As shown in Table 1, matched text alone enables high accuracy (e.g., GPT-4o: 90.07%), confirming it carries answer-relevant information. Critically, corrupted text yields low text-only accuracy (GPT-4o: 17.59%; Claude Sonnet: 16.17%), confirming that the corrupted descriptions are misleading and do not coincidentally lead to correct answers. Irrelevant text similarly yields low accuracy (GPT-4o: 18.67%), verifying its irrelevance. This validates that performance drops under corruption are attributable to models trusting misleading text over visual evidence, rather than to artifacts in the text construction.

	VQAv2			
Model	Base	Match	Corruption	Irrelevance
Claude Sonnet	66.88	84.39	16.17	24.39
GPT-4o	78.39	90.07	17.59	18.67
Molmo-7B-D	76.33	88.98	18.74	35.40

Table 1. Text-only accuracy (%) across different models. It provides a sanity check for the constructed text when matched, corrupted, or irrelevant.

3. Empirical Analysis

We now present results across all four datasets and ten models, focusing on how corrupted text affects model behavior. Table 2 reports base accuracy, corruption accuracy, normalized accuracy, and TPR for VQAv2, DocVQA, and MathVista; Table 3 presents the corresponding results for brand detection. Figure 2 visualizes the distribution of image-consistent, text-consistent, and other behaviors across models and text conditions, while Figure 3 shows the TPR of all models under different text variations on VQAv2.

3.1. Text Bias Under Corruption

We first examine model behavior under the corrupted text condition, which directly probes whether VLMs follow misleading text over clear visual evidence.

The results in Table 2 reveal a stark divide between proprietary and open-weight models. Among proprietary systems, Claude Sonnet exhibits exceptional robustness, achieving a TPR of only 9.58% on VQAv2 and 3.21% on DocVQA—its normalized accuracy even exceeds 100%, indicating that corrupted text occasionally helps the model reconsider borderline answers. GPT-4o also demonstrates strong resilience, with a VQAv2 TPR of 27.09% and a normalized accuracy of 90.25%.

In contrast, open-weight models overwhelmingly defer to corrupted text. LLaVA-NeXT-7B exhibits a TPR of 85.52% on VQAv2 and 87.77% on DocVQA, with normalized accuracy plummeting to 36.10% and 18.60%, respectively. Even the larger LLaVA-NeXT-34B variant maintains a TPR above 67% on VQAv2. Phi3.5 (TPR 74.05%) and Molmo-7B-D (TPR 59.40%) follow a similar pattern. These models are effectively "reading" the corrupted text rather than "seeing" the image.

A notable exception among open-weight models is Qwen2-VL-7B, which achieves a VQAv2 TPR of only 29.22%—comparable to GPT-4o—despite its relatively modest parameter count. This suggests that architectural and training choices can substantially mitigate text bias independently of model scale.

On MathVista, even proprietary models show increased vulnerability: GPT-4o's TPR rises to 48.98%, and LLaVA-NeXT-7B reaches 84.19%. This indicates that when visual evidence requires multi-step reasoning, models are more susceptible to textual shortcuts.

3.2. Brand Detection: Real-World Risk

Brand detection represents a safety-critical application where text bias could enable adversarial manipulation—for instance, overlaying text on a product image to cause brand misidentification. Table 3 presents results on this task.

The proprietary–open divide is even more pronounced here. GPT-4o and Claude Sonnet are virtually immune to text corruption, with TPRs of only 0.83% and 0.96%, respectively—both achieving normalized accuracies above 100%. These models reliably identify brands from visual features regardless of conflicting textual cues.

Open-weight models, however, exhibit alarming vulnerability. Molmo-7B-D drops to a normalized accuracy of only 47.39% with a TPR of 60.40%, meaning that more than half the time it follows the corrupted text over clear visual brand indicators. LLaVA-NeXT-7B (TPR 59.17%) and Phi3.5 (TPR 50.45%) show similar susceptibility. These failure rates have direct security implications: an attacker could potentially manipulate brand identification in de-

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Model	VQAv2				DocVQA				MathVista			
	Base ↑	Corruption ↑	Norm ↑	TPR ↓	Base ↑	Corruption ↑	Norm ↑	TPR ↓	Base ↑	Corruption ↑	Norm ↑	TPR ↓
GPT-4o mini	69.82	51.55	73.83	52.42	69.40	38.20	55.04	52.07	52.30	23.90	45.70	80.28
Claude Haiku	50.08	25.54	50.99	82.70	68.80	40.20	58.43	47.67	41.00	19.80	48.29	77.42
GPT-4o	78.39	**70.75**	90.25	27.09	85.00	73.60	86.59	17.96	**58.90**	41.20	69.95	48.98
Claude Sonnet	66.88	68.17	**101.93**	**9.58**	87.00	**84.60**	**97.24**	**3.21**	56.30	**49.30**	**87.57**	**29.14**
LLaVA-NeXT-7B	79.45	28.69	36.10	85.52	53.60	10.00	18.60	87.77	35.80	19.70	54.97	84.19
LLaVA-NeXT-13B	81.02	37.61	46.40	74.43	57.70	11.00	19.10	86.84	36.20	20.60	56.89	80.83
LLaVA-NeXT-34B	82.96	42.87	51.70	67.56	64.00	15.10	23.61	82.69	34.00	21.70	61.98	67.64
Phi3.5	75.65	35.23	46.50	74.05	78.20	50.50	64.60	40.51	43.10	22.20	51.47	80.20
Molmo-7B-D	76.33	49.29	64.50	59.40	74.00	38.40	51.90	57.20	44.90	32.90	73.27	60.63
Qwen2-VL-7B	**85.51**	50.79	59.41	29.22	**90.50**	57.50	63.63	37.41	55.40	28.90	52.18	70.23

Table 2. Performance (%) reported as Base Accuracy, Corruption Accuracy, Normalized Corruption Accuracy (Norm) and Text Preference Ratio (TPR) under corruption. **Bold**: best performance; underline: second best. Full results under all text variations are in the Appendix.

ployed systems simply by injecting misleading text into the model's context.

Model	Brand Detection			
	Base ↑	Corruption ↑	Norm ↑	TPR ↓
GPT-4o mini	88.84	84.8	95.44	7.48
Claude Haiku	84.40	78.72	93.27	6.44
GPT-4o	88.68	**89.76**	101.22	**0.83**
Claude Sonnet	**90.20**	90.24	100.04	0.96
LLaVA-NeXT-7B	78.60	55.32	70.39	59.17
LLaVA-NeXT-13B	83.00	60.00	72.29	40.65
LLaVA-NeXT-34B	66.28	53.52	80.77	23.49
Phi3.5	84.40	60.68	71.90	50.45
Molmo-7B-D	87.44	41.44	47.39	60.40
Qwen2-VL-7B	89.68	86.48	96.43	2.99

Table 3. Performance on the Brand Detection dataset reported in Base Accuracy, Corruption Accuracy, Normalized Corruption Accuracy (Norm), and Text Preference Ratio (TPR). **Bold**: best performance; underline: second best performance.

3.3. Matched and Irrelevant Text

When text matches the visual ground truth (*matched* condition), all models generally improve, with match accuracies consistently exceeding base accuracies (e.g., LLaVA-NeXT-7B rises from 79.45% to 92.32% on VQAv2; see Table 4). This confirms that models do integrate textual information and can benefit from accurate supplementary text.

Under the *irrelevant* text condition, the impact is more nuanced. Most models maintain accuracy close to their base performance (e.g., LLaVA-NeXT-7B: 79.43% vs. 79.45% base on VQAv2), indicating that they can largely ignore clearly unrelated text. However, as we show in Section 4.4, seemingly relevant but factually uninformative text produces significantly more bias than obviously unrelated text, revealing that models are sensitive to superficial topical overlap rather than genuine informational content. This finding has practical implications for RAG-augmented deployments, where retrieved passages often share domain vocabulary with the query image yet may contain factually incorrect details. System designers should account for this heightened risk when retrieved text is topically plausible but factually unreliable. We further observe that among open-weight models, Qwen2-VL-7B is uniquely resistant to irrelevant text, dropping fewer than 0.5 percentage points on VQAv2 relative to its base accuracy—suggesting its training regime has instilled a stronger visual grounding prior that resists distraction even when topically related noise is present.

3.4. Cross-Dataset Patterns

Comparing results across all four datasets reveals several consistent regularities. First, the proprietary–open-weight divide is robust across task types: proprietary models maintain low TPRs regardless of whether the task requires general perception, document understanding, or mathematical reasoning. Second, task complexity moderates text bias even among robust models—GPT-4o's TPR grows from 27.09% on VQAv2 to 48.98% on MathVista, indicating that multi-step visual reasoning increases susceptibility to textual shortcuts. Third, open-weight models exhibit relatively stable vulnerability across tasks, with LLaVA-NeXT-7B sustaining TPRs above 80% on both VQAv2 and DocVQA, suggesting that text preference reflects a fixed characteristic of each model's modality integration rather than task-specific behavior. Together, these cross-dataset regularities confirm that text bias is a systematic property of a model's architecture and training rather than an artifact of any particular evaluation domain.

Across all tasks, we also observe that irrelevant text has negligible impact on model behavior compared to corrupted text, with accuracy differences between irrelevant and base conditions remaining below 2 percentage points for most models. This confirms that it is the semantic conflict—rather than mere textual length or noise—that drives modality bias. Notably, the task-specific difficulty of constructing plausible corruptions may partly explain why Brand Detection elicits lower TPRs even among open-weight models: brand names and visual indicators are highly constrained domains, making it harder for corrupted text to offer credibly misleading alternatives that a model would adopt over what is visually apparent in the screenshot.

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4. Investigated Solutions

4.1. Prompt Instructions

A natural first mitigation is to explicitly instruct the model to prioritize visual evidence. We prepend a cautionary instruction ("The provided text may be inaccurate. When in doubt, trust the image over the text.") to the prompt. As shown in Figure 4, this marginally reduces TPR—for example, Qwen2-VL-7B's TPR decreases by a few percentage points under instruction. However, the effect is limited: instruction alone is insufficient to overcome deeply learned text priors, particularly for models with high baseline text bias. Indeed, for LLaVA-NeXT-7B, the cautionary instruction decreases TPR by fewer than 5 percentage points while base accuracy remains largely unchanged, confirming that inference-time prompting cannot compensate for the modality imbalances encoded in model weights during pretraining.

4.2. Language Model Size

We examine whether scaling the language model backbone reduces text bias by comparing LLaVA-NeXT at 7B, 13B, and 34B parameters. As shown in Figure 5, increasing model size monotonically decreases TPR on VQAv2: from 85.52% (7B) to 74.43% (13B) to 67.56% (34B). While this is a meaningful reduction, even the 34B model retains a TPR exceeding 67%, indicating that scale alone does not resolve the fundamental text bias. Larger language models may develop better calibration and reasoning capabilities, but the dominance of textual priors persists.

4.3. Token Order

The position of textual tokens relative to image tokens in the input sequence may influence modality preference. We evaluate Phi3.5 with text tokens placed before vs. after image tokens. As shown in Figure 6, placing text before the image increases TPR compared to the text-after-image configuration. This is consistent with the autoregressive nature of the language model decoder and with documented "lost-in-the-middle" effects [25]: earlier tokens exert stronger influence on subsequent generation, and text placed first may establish a prior that the model struggles to override when processing the image.

4.4. Text Relevance

Not all irrelevant text is equally distracting. We compare topically related but factually uninformative text versus clearly unrelated text drawn from random Wikipedia passages. As shown in Figure 7, seemingly relevant irrelevant text produces higher TPR than obviously unrelated text. This indicates that models are susceptible to superficial topical similarity—text that "looks like" it could be relevant triggers stronger integration with visual processing, even when it provides no useful information.

	VQAv2				
Model	Base ↑	Match ↑	Corruption ↑	Irrelevance ↑	Macro ↑
LLaVA-NeXT-7B	**79.45**	**92.32**	28.69	**79.43**	66.81
Instruction	**79.45**	92.25	34.27	78.15	68.22
SFT	77.48	87.56	**71.25**	77.32	**78.71**
Qwen2-VL-7B	**85.51**	**92.76**	50.79	83.70	75.75
Instruction	**85.51**	92.62	54.78	82.82	76.74
SFT	84.18	87.01	**82.72**	**84.00**	**84.58**

Table 4. In-distribution performance comparison between original models, instruction and fine-tuned models.

4.5. Uni-modal Certainty

We investigate whether a model's confidence on the image alone predicts its susceptibility to text corruption. For each sample, we measure the model's accuracy under image-only input and compare its behavior when corrupted text is added. Figure 8 and Figure 9 illustrate this relationship for GPT-4o and LLaVA-NeXT-7B, respectively. Models that answer correctly with high confidence from the image alone are more likely to resist corrupted text, while samples where the model is uncertain from visual evidence alone are disproportionately flipped by conflicting text. This suggests that text bias is most dangerous precisely when visual evidence is ambiguous—the setting where a robust model should seek additional verification rather than defaulting to text. Concretely, for LLaVA-NeXT-7B on VQAv2, when the model fails on the image-only input, adding corrupted text induces text-following behavior in over 94% of cases; when the model is correct image-only, this rate drops to 78%—still alarmingly high, but confirming that stronger visual grounding does provide partial protection against misleading text.

4.6. Supervised Fine-tuning

Given the limited effectiveness of the preceding approaches, we turn to supervised fine-tuning (SFT) with text-augmented training data as a more direct intervention. We construct training sets that pair images with all three text conditions (matched, corrupted, irrelevant) and fine-tune LLaVA-NeXT-7B and Qwen2-VL-7B on VQAv2 training data augmented in this manner.

In-distribution results. Table 4 shows that SFT dramatically improves corruption accuracy: LLaVA-NeXT-7B increases from 28.69% to 71.25%, and Qwen2-VL-7B from 50.79% to 82.72%. The macro-averaged accuracy across all conditions also improves substantially (LLaVA-NeXT-7B: 66.81% → 78.71%; Qwen2-VL-7B: 75.75% → 84.58%). The cost is a modest drop in base accuracy of approximately 1–2 percentage points, an acceptable trade-off for substantially improved robustness.

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Model	DocVQA		MathVista		Brand Detection	
	Base ↑	Macro ↑	Base ↑	Macro ↑	Base ↑	Macro ↑
LLaVA-NeXT-7B	**53.60**	51.07	**35.80**	41.03	78.60	46.44
Instruction	**53.60**	49.27	**35.80**	41.20	78.60	47.36
SFT	52.20	**56.17**	35.30	**41.63**	**81.36**	**72.29**
Qwen2-VL-7B	**90.50**	80.83	55.40	53.87	**89.68**	81.85
Instruction	**90.50**	80.77	55.40	54.10	**89.68**	84.48
SFT	90.30	**88.97**	**58.50**	**57.17**	89.44	**88.75**

Table 5. Performance comparison with Base and Macro accuracy based on DocVQA, MathVista, and Brand Recognition. See full results under different text conditions in Appendix.

Out-of-distribution generalization. Crucially, the benefits of SFT generalize to held-out tasks not seen during fine-tuning. As shown in Table 5, LLaVA-NeXT-7B's macro accuracy improves on DocVQA (51.07% → 56.17%), MathVista (41.03% → 41.63%), and Brand Detection (46.44% → 72.29%). Qwen2-VL-7B shows even stronger generalization: DocVQA (80.83% → 88.97%), MathVista (53.87% → 57.17%), and Brand Detection (81.85% → 88.75%). These results indicate that SFT teaches models a generalizable strategy for arbitrating between modalities rather than merely memorizing task-specific patterns.

Data ablation. Figure 10 presents an ablation study on the amount of SFT data required. Performance improves rapidly with even small amounts of augmented training data and plateaus after a moderate quantity, suggesting that the approach is data-efficient and practical for deployment. Importantly, the quality of text augmentations matters: GPT-4o-generated corruptions, which are semantically coherent and plausible, yield stronger fine-tuned models than naive heuristic corruptions such as random label substitution, underscoring that exposure to realistic, meaningful misleading text is critical for developing robust modality arbitration.

5. Theoretical Analysis

We provide a theoretical account of why text bias arises in VLMs. Consider a VLM that, given an image I and text T, produces an answer by implicitly computing:

$$P(a \mid I, T) \approx \alpha \cdot P(a \mid T) + (1 - \alpha) \cdot P(a \mid I), \quad (1)$$

where $\alpha \in [0, 1]$ represents the model's effective reliance on the textual modality. This mixture view, while simplified, captures the core tension: when α is high, the model defers to text even when visual evidence is unambiguous. More precisely, α encodes the model's implicit confidence in each modality: a well-calibrated VLM should reduce α when textual information conflicts with high-confidence visual signals. The observed text bias suggests that most open-weight models lack this adaptive calibration, applying a fixed, text-heavy weighting regardless of visual certainty.

The parameter α is shaped by training data composition. Modern VLMs are built on language model backbones pretrained on trillions of text-only tokens [2, 39], whereas vision-language alignment is typically performed on orders of magnitude less paired data [23, 50]. This imbalance means that the language model's prior $P(a \mid T)$ is far more confident and well-calibrated than the visual posterior $P(a \mid I)$, leading to a high effective α.

Empirically, we observe that models with stronger language model priors—as measured by text-only accuracy on our benchmark—exhibit higher TPR. The LLaVA-NeXT family provides a controlled comparison: scaling from 7B to 34B parameters improves the language backbone's capacity and correspondingly increases base accuracy, yet TPR decreases only modestly (85.52% → 67.56%). This is consistent with larger models developing better visual grounding that partially counteracts the text prior, while the overall dominance of the textual pathway remains.

Formally, let $r = D_V/(D_T + D_V)$ denote the fraction of vision-language data in pretraining, where D_T and D_V are the volumes of text-only and paired image-text examples, respectively. Under a simplified maximum-likelihood model, the learned modality weight satisfies $\hat{\alpha} \approx 1 - r$, since the language model prior is calibrated primarily from text. When $D_T \gg D_V$—as is typical in current VLMs [39]—we have $r \ll 1$, yielding $\hat{\alpha} \approx 1$ and strong text bias. This analysis predicts that substantially increasing the proportion of vision-language data in pretraining would reduce $\hat{\alpha}$ proportionally, suggesting that future foundation models with more balanced training corpora may exhibit lower text bias without task-specific fine-tuning.

SFT with text-augmented data can be understood as directly training the routing mechanism to reduce α when visual and textual signals conflict. By exposing the model to corrupted text during training, we teach it to down-weight $P(a \mid T)$ in favor of $P(a \mid I)$ when inconsistencies are detected. The out-of-distribution generalization observed in Table 5 suggests that this recalibration operates at the level of modality arbitration rather than task-specific pattern matching. This framework additionally clarifies why prompt instructions offer only marginal relief: operating purely at inference time, they cannot alter the learned weight distributions that encode the model's text prior. SFT, by contrast, directly reshapes the parameters governing modality arbitration, yielding durable reductions in α that persist across domains.

An important corollary of this analysis concerns the design of vision-language alignment data. Our framework suggests that the ratio r can be effectively increased not only by adding more paired image-text examples in pretraining, but also by constructing training scenarios that explicitly reward the model for overriding strong textual priors when they conflict with visual evidence. This is

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precisely what our SFT approach achieves: each training step on a corrupted-text example applies a gradient signal that locally reduces α, encouraging the model to rely on $P(a \mid I)$ even when $P(a \mid T)$ is highly confident. The data-efficient plateau observed in Figure 10 is consistent with this view—once the model has seen a sufficient number of conflict examples to recalibrate its modality weighting, additional examples yield diminishing returns. This observation suggests that targeted augmentation with high-quality, semantically coherent corrupted examples may be substantially more parameter-efficient than large-scale pretraining on additional image-text pairs as a strategy for mitigating text bias in deployed VLMs.

6. Related Work

VQA and multimodal benchmarks. Visual question answering has served as a primary testbed for vision-language understanding since the introduction of VQAv2 [12] and its extensions to document understanding [28], mathematical reasoning [27], and knowledge-intensive settings [32]. More recent benchmarks such as MMBench [26], MMMU [46], and SEED-Bench [20] evaluate broad multimodal capabilities. However, these benchmarks assume cooperative inputs where all modalities align toward the correct answer, and they do not probe robustness under modality conflicts. BFiT fills this gap by systematically introducing controlled text–image disagreements.

Hallucination and reliability in VLMs. A growing body of work studies hallucination in VLMs, where models generate outputs not grounded in the visual input [13, 21, 31]. Object hallucination benchmarks [4, 33], decoding-time interventions [6, 15, 18], and post-hoc correction methods [5, 41, 45] have been proposed. While hallucination work focuses on generation unfaithful to the image, BFiT studies a complementary failure mode: models faithfully following *text* that contradicts the image, revealing a systematic modality preference rather than a generation artifact.

Text and modality bias. Prior work has documented language priors and shortcut biases in VQA models [12, 48] and shown that large language models are susceptible to irrelevant context [34, 35]. Studies on compositional understanding reveal that contrastive models often rely on textual "bags of words" rather than visual structure [14, 37, 47]. Concurrent work by Deng et al. [8] and Tong et al. [38] examines visual grounding failures. Cross-modality knowledge conflict in VLMs is studied by Zhu et al. [51], and the broader phenomenon of LLMs overriding internal priors with external evidence is quantified by Wu et al. [43] and Xie et al. [44]. BFiT extends these lines of inquiry by providing a controlled framework that independently varies text quality and measures its causal effect on model predictions across diverse vision tasks.

Retrieval-augmented and agent-based VLMs. Retrieval-augmented generation [19, 36] and verification frameworks [10, 11, 30] aim to improve factuality by grounding models in retrieved evidence. VLMs are increasingly embedded in agentic pipelines for web navigation [17, 49] and GUI interaction [16], where noisy or adversarial text is common. Our benchmark directly simulates these realistic conditions by pairing images with unreliable textual context, exposing vulnerabilities that purely visual evaluations miss.

7. Conclusion and Discussion

We introduced BFiT, a diagnostic benchmark for evaluating how VLMs arbitrate between conflicting visual and textual evidence across four diverse tasks. Our evaluation of ten models reveals that text bias is pervasive among open-weight VLMs, with models such as LLaVA-NeXT-7B exhibiting TPRs exceeding 85% on VQAv2 and DocVQA. Proprietary models, particularly Claude Sonnet and GPT-4o, demonstrate substantially greater robustness. We systematically investigated factors influencing text bias—model scale, token order, text relevance, uni-modal certainty, and prompt instructions—finding that while each factor has measurable effects, none fully resolves the problem. Supervised fine-tuning with text-augmented data emerged as the most effective mitigation, dramatically improving corruption robustness while generalizing to out-of-distribution tasks.

Limitations. Our benchmark evaluates only English-language tasks; text bias may manifest differently in multilingual settings. The SFT approach requires labeled data with controlled text conditions, which adds annotation overhead. Additionally, our corruption strategy uses plausible but synthetically generated incorrect text, which may not fully capture the diversity of real-world textual noise. Relatedly, our SFT pipeline relies on GPT-4o to produce training corruptions, introducing a dependency on proprietary APIs; future work should explore self-supervised corruption strategies that do not require external oracles, making the mitigation more broadly accessible. Furthermore, evaluating text bias in multi-turn dialogue settings—where contextual cues accumulate across turns—remains an important open direction, as sequential context may amplify modality imbalances in ways not captured by single-turn protocols.

Future directions. Promising avenues include adversarial text injection attacks that exploit text bias in deployed

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systems, development of multimodal faithfulness metrics that measure modality arbitration quality, and extension to video and multi-turn dialogue settings where temporal context introduces additional complexity. Investigating the internal attention mechanisms that drive text bias—for example, whether cross-modal attention heads systematically upweight textual tokens—could yield targeted architectural interventions beyond data-level SFT, offering a more principled path toward robust modality arbitration.

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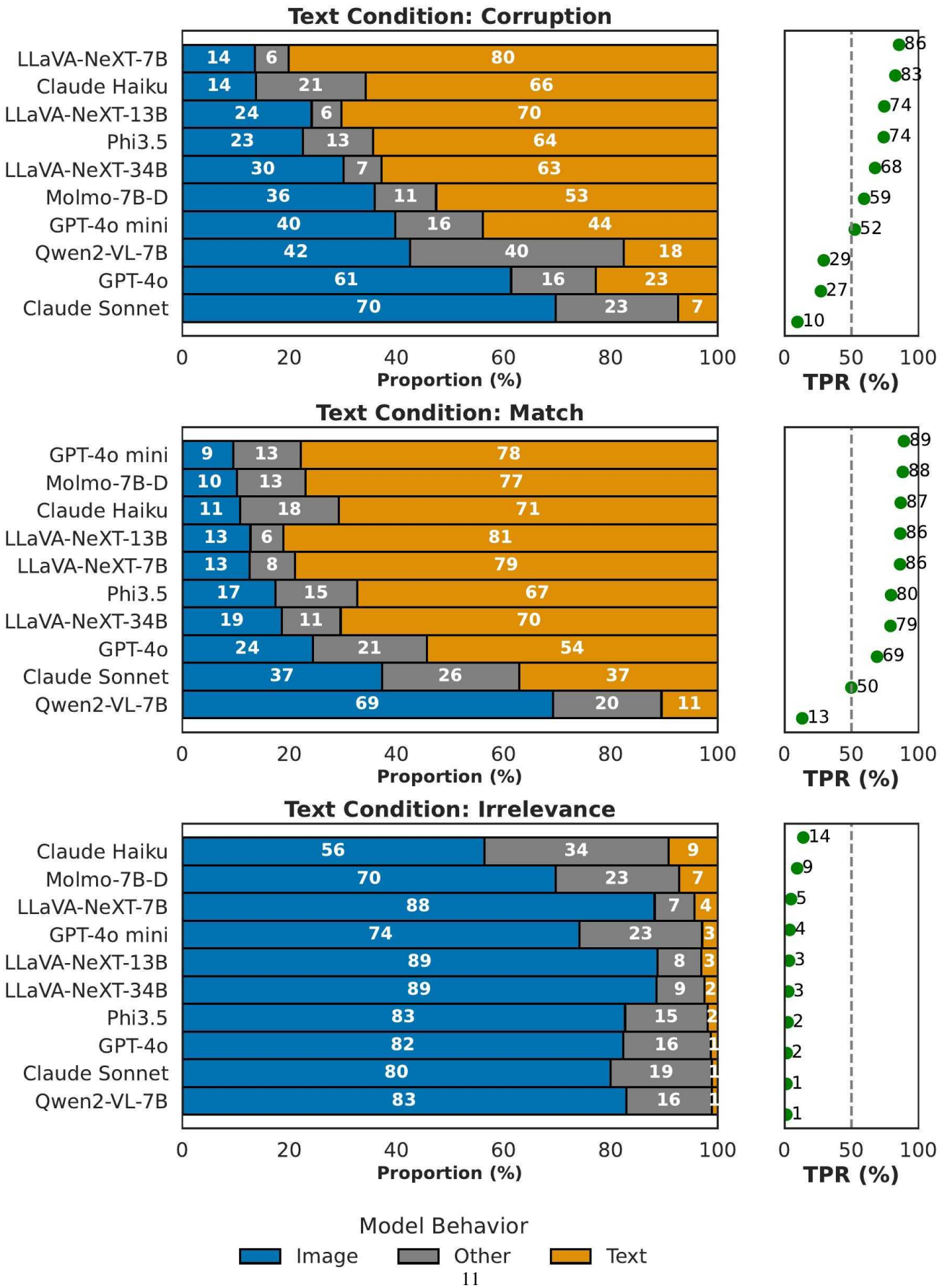

Figure 2. Distribution of model prediction behaviors (image-consistent, text-consistent, other) across all models under matched, corrupted, and irrelevant text conditions.

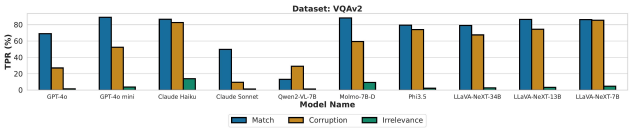

Figure 3. Text Preference Ratio (TPR) of all models under different text variations on VQAv2. Open-weight models exhibit substantially higher TPR under corruption than proprietary models.

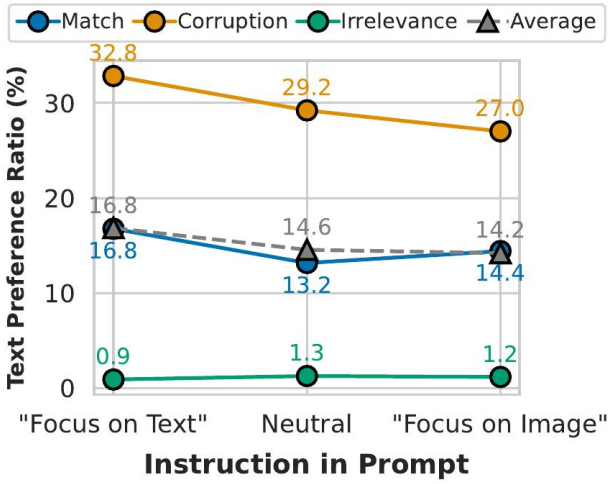

Figure 4. Effect of cautionary prompt instructions on TPR for Qwen2-VL-7B across text conditions. Instructions produce only modest reductions in text preference.

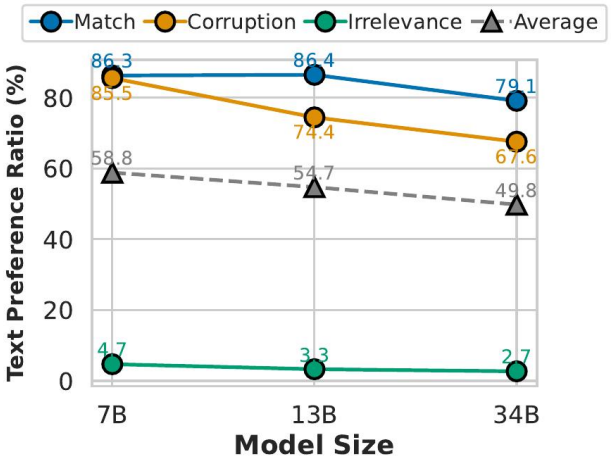

Figure 5. Effect of language model backbone size on TPR for the LLaVA-NeXT family. Larger models show reduced but still substantial text bias.

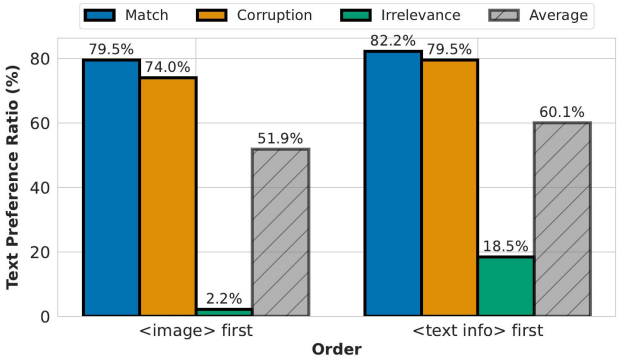

Figure 6. Effect of token ordering on text bias for Phi3.5. Placing text before image tokens increases the TPR.

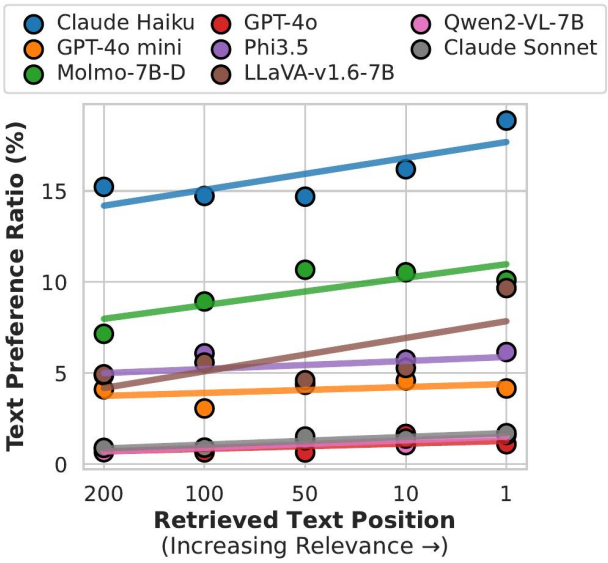

Figure 7. Effect of text relevance on TPR. Seemingly relevant but uninformative text induces greater bias than clearly unrelated text.

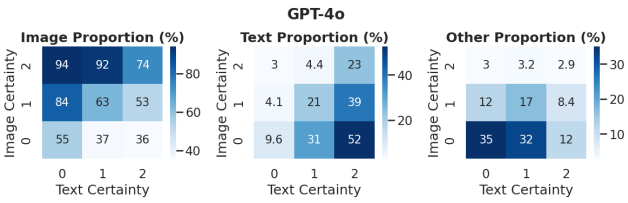

Figure 8. Relationship between image-only prediction certainty and susceptibility to text corruption for GPT-4o.

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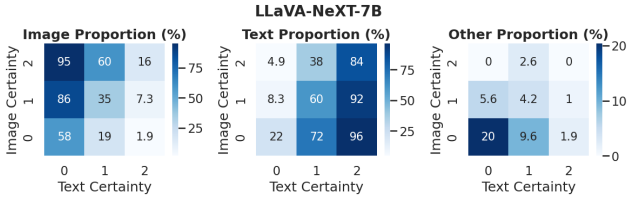

Figure 9. Relationship between image-only prediction certainty and susceptibility to text corruption for LLaVA-NeXT-7B.

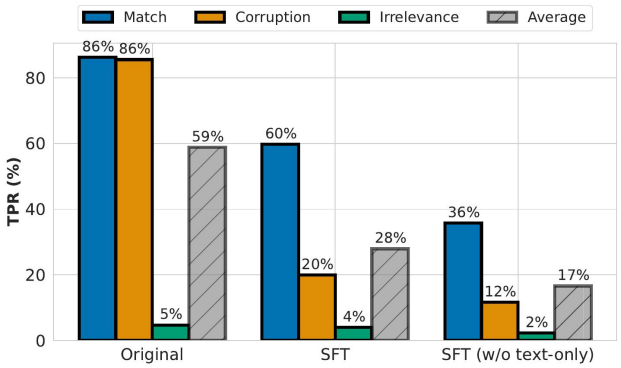

Figure 10. Ablation on the amount of SFT training data for LLaVA-NeXT-7B. Corruption accuracy improves rapidly and plateaus, indicating data efficiency.

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